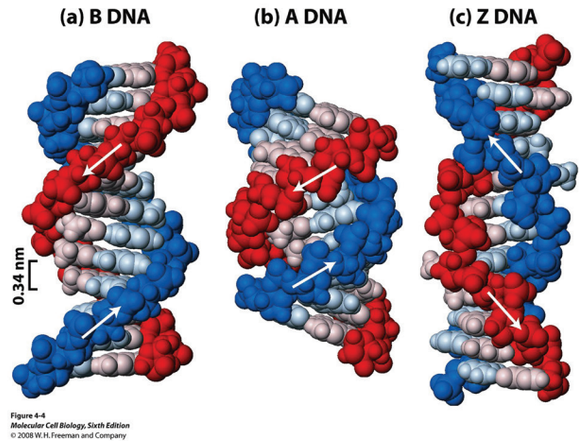


Figure 14.10: Forms of DNA

HOTS

Write the base sequence of complementary DNA and RNA strand for the following.

5′GCGCAATATTTCT3′

14.3 DNA Replication

DNA is a marvelous device for the stable storage of genetic information. DNA replication is a process in which copies of DNA molecules are faithfully made. Here double stranded DNA molecule is copied to produce two identical dsDNA molecules. Replication is an essential process because whenever a cell divides, the two new daughter cells must contain the same genetic information in DNA as parent cell. DNA replication occurs during the S (DNA synthesis) phase and precedes cell division.

Watson and Crick proposed the hypothesis of semiconservative replication. According to them each DNA strand serves as a template for the synthesis of a new strand, producing two new DNA molecules each with one old strand and one new strand. (Figure 14.11).

Infobits

Max Delbruck suggested that there could be three possible ways in which DNA could replicate.

Semiconservative – DNA replication that produce two copies of double stranded DNA (dsDNA) each containing one old strand and one new strand.

Conservative – DNA replication that produces two daughter ds DNAs, one of which consists of two original strands whereas the other daughter DNA consists of two newly synthesised strands.

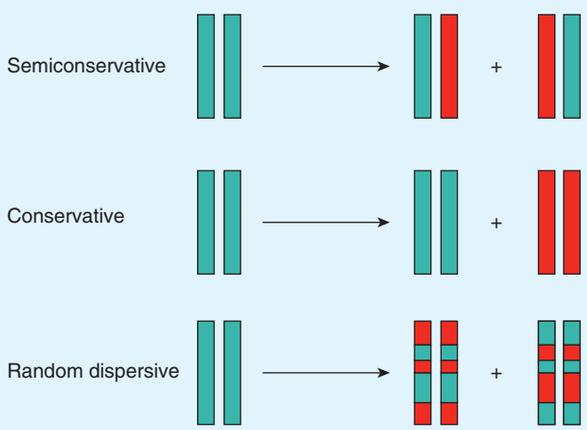


Figure 14.11: Semiconservative, conservative and dispersive replication

Dispersive – DNA replication in which the original dsDNA undergoes fragmentation, the fragments synthesize complementary structure both of which assemble to form two replicas.

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